

NORTHWIND SALES ANALYSIS

Executive Business Analysis

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Tools: MySQL, SQL, LibreOffice

Project Type: Sales & Business Performance Analysis

Project Overview

This project analyzes the Northwind sales database to evaluate overall sales performance, revenue trends, customer behavior, product performance, employee performance and key business risks.

SQL was used to transform the raw relational data into business-focused metrics and insights. The analysis combines transactional, customer, product, category and employee data to provide an integrated view of business performance.

Business Objectives

The analysis was designed to answer the following business questions:

1. How much revenue did the business generate?
2. How many orders and customers were involved?
3. How did revenue change over time?
4. Which customers generate the most revenue?
5. Which products and categories perform best?
6. Which employees generate the most sales?
7. How strong is customer retention?
8. How concentrated is revenue among major customers?
9. Are there significant data-quality issues?
10. What actions could management take based on the findings?

Dataset Overview

Dataset Component	Records/Items
Customers	89
Orders	830
Order Details	2,155
Products	77
Categories	8

Dataset Component Records/Items

Employees 9

The analysis covers sales transactions across customers, orders, order details, products, categories and employees. Revenue was calculated from order-detail records using unit price, quantity and recorded discount information.

Executive KPI Summary

KPI	Result
Total Revenue	1,354,458.59
Total Orders	830
Active Customers	89
Products Sold	77
Active Employees	9
Units Sold	51,317
Average Order Value	1,631.88
Repeat Customer Rate	98.9%
Top 10 Customer Revenue Share	45.5%
Average Recorded Discount	0%

The business generated 1,354,458.59 in revenue across 830 orders and 51,317 units sold. Customer retention was particularly strong, with 88 of 89 active customers classified as repeat customers.

Revenue Trend Analysis

Year Revenue

2013 226,298.50

2014 658,388.75

2015 469,771.34

Revenue increased substantially between 2013 and 2014, rising from 226,298.50 to 658,388.75. Revenue subsequently declined to 469,771.34 in 2015.

The 2014 peak followed by the 2015 decline represents one of the most important performance patterns identified in the analysis and warrants further investigation into the underlying customer, product, geographic and sales-representative drivers.

Customer Analysis

Customer	Country	Revenue
QUICK-Stop	Germany	117,483.39
Save-a-lot Markets	USA	115,673.39
Ernst Handel	Austria	113,236.68
Hungry Owl All-Night Grocers	Ireland	57,317.39
Rattlesnake Canyon Grocery	USA	52,245.90

The customer analysis indicates strong customer loyalty, with a 98.9% repeat-customer rate.

However, the top 10 customers account for 45.5% of total revenue. This creates a degree of key-account dependency and means that customer retention should remain a strategic priority.

Product & Category Analysis

Product	Category	Units Sold	Revenue
Côte de Blaye	Beverages	623	149,984.20
Thüringer Rostbratwurst	Meat & Poultry	746	87,736.40
Raclette Courdavault	Dairy Products	1,496	76,296.00
Camembert Pierrot	Dairy Products	1,577	50,286.00
Tarte au sucre	Confections	1,083	49,827.90

Côte de Blaye was the strongest revenue-generating product, producing 149,984.20. The difference between unit volume and revenue among leading products demonstrates why both measures should be considered when evaluating product performance.

Employee Performance

Employee	Orders	Revenue
Margaret Peacock	156	250,187.45
Janet Leverling	127	213,051.30
Nancy Davolio	123	202,143.71
Andrew Fuller	96	177,749.26
Robert King	72	141,295.99
Laura Callahan	104	133,301.03
Anne Dodsworth	43	82,964.00
Michael Suyama	67	78,198.10
Steven Buchanan	42	75,567.75

Margaret Peacock generated the highest recorded employee revenue at 250,187.45. Employee performance varied substantially across the nine active sales representatives.

These differences should be investigated alongside customer portfolios, territories, order volumes and product mix before attributing performance differences solely to individual sales effectiveness.

Discount Analysis

No discounts were recorded in the supplied order-detail dataset. All 2,155 order-detail records contained a discount value of 0, with no NULL discount values.

Therefore, the analysis records an average discount of 0% and an estimated recorded discount value of 0.

Data Quality & Validation

Multiple validation checks were performed to assess the reliability of the analysis. These included checks for missing customer and employee references, orphan order-detail records, duplicate identifiers, invalid quantities or prices, missing product categories, date ranges and revenue reconciliation.

The final revenue reconciliation successfully matched, confirming consistency between the revenue calculations used in the analysis.

Key Business Findings

Finding 1 — Strong overall performance

The business generated **1,354,458.59** in revenue from **830 orders**.

Finding 2 — Exceptional customer retention

98.9% of active customers were repeat customers.

Finding 3 — Revenue concentration

The top 10 customers contributed **45.5%** of total revenue.

Finding 4 — Revenue volatility

Revenue peaked at **658,388.75 in 2014** before declining to **469,771.34 in 2015**.

Finding 5 — Product performance is not simply volume-driven

Côte de Blaye generated **149,984.20** despite selling fewer units than some other leading products.

Finding 6 — Sales performance varies significantly

Margaret Peacock generated **250,187.45**, the highest employee revenue in the analysis.

Business Risks**Risk 1 — Key-account dependency**

45.5% of revenue comes from the top 10 customers.

Risk 2 — Revenue slowdown

The 2015 decline requires investigation.

Risk 3 — Employee performance concentration

Revenue is not evenly distributed across employees.

Risk 4 — Limited discount information

Although discounts are recorded as zero, the dataset cannot tell us whether promotional activity existed outside the recorded field.

Recommendations

1. Protect key accounts

Prioritize retention strategies for the top revenue-generating customers.

2. Investigate the 2015 decline

Break the decline down by:

- month
- customer
- product
- category
- employee
- country

3. Protect high-value products

Monitor Côte de Blaye and other high-revenue products for availability and demand.

4. Investigate employee performance differences

Compare representatives by:

- orders
- revenue
- customers
- product mix
- average order value

5. Expand the customer base

Maintain the excellent repeat-customer rate while acquiring additional customers.

6. Improve promotional data capture

If discounts or promotions exist operationally, ensure they are properly recorded in the database.

Conclusion

Use:

The Northwind sales analysis demonstrates a business with strong revenue generation, exceptional repeat-customer behavior and several high-performing products and sales representatives. However, the analysis also identifies important management

considerations, particularly revenue concentration among major customers and the significant decline following the 2014 revenue peak.

The findings demonstrate how SQL can be used not only to retrieve data but to transform transactional records into actionable business intelligence. The resulting analysis provides management with a structured basis for customer retention, sales performance monitoring, product prioritization and future revenue-growth strategies.

Technical Skills Demonstrated

MySQL

SQL

CTEs

JOINS

Aggregate Functions

Window Functions

CASE Statements

Subqueries

Data Validation

Revenue Analysis

Customer Analysis

Product Analysis

Employee Performance Analysis

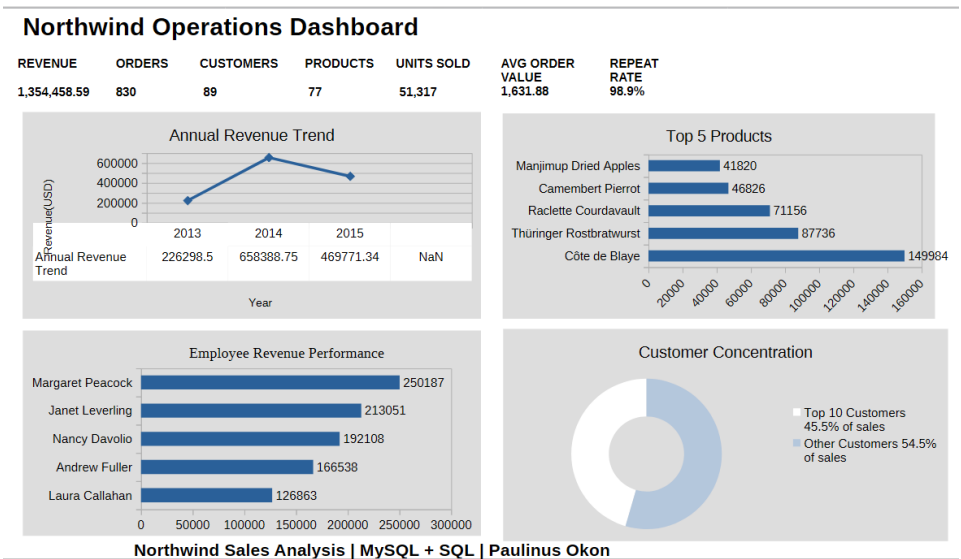
Business Intelligence

Dashboard Development

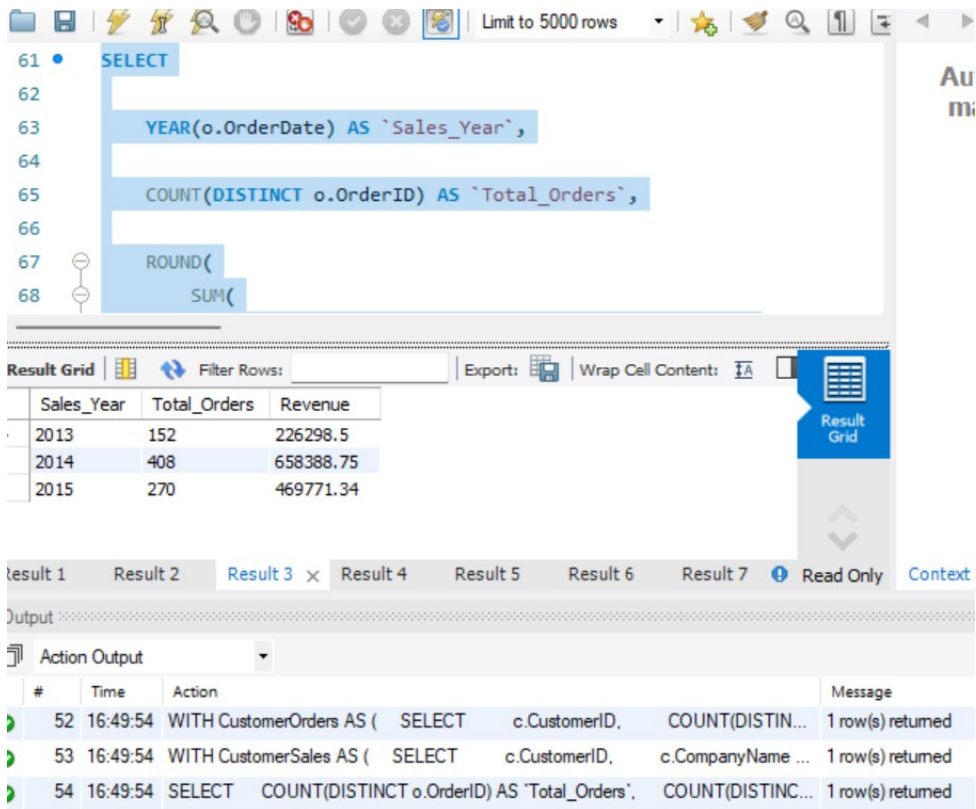
Data Storytelling

OUTPUT IMAGES

1. Executive KPI/Dashboard section



2. Revenue Trend section



3. Customer/Product section

Top Customers	Revenue	Top Products	Revenue
QUICK-Stop	117483.39	Côte de Blaye	149984.2
Save-a-lot Markets	115673.39	Thüringer Rostbratwurst	87736.4
Ernst Handel	113236.68	Raclette Courdavault	76296
Hungry Owl	57317.39	Camembert Pierrot	50286
Rattlesnake Canyon	52245.9	Tarte au sucre	49827.9

4. Data Quality section

1	Validation Check	Result
2	Orphan Order Details	0
3	Duplicate Order IDs	0
4	Duplicate Product IDs	0
5	Invalid Quantity/Price/Discount	0
6	Products Without Category	0
7	NULL Discounts	0
8	Revenue Reconciliation	PASS
9	Total Order Details	2155
10	Minimum Discount	0
11	Maximum Discount	0
12	Average Discount	0
13	NULL Discount Count	0

Key Findings at a Glance

REVENUE

1,354,458.59

Total revenue generated

REVENUE PEAK

658,388.75

2014 revenue

Revenue Concentration

45.5%

Top 10 customers' share

CUSTOMER RETENTION

98.9%

Repeat customer rate

Top Employee

Margaret Peacock

250,187.45 revenue

Top Product

Côte de Blaye

149,984.20 revenue